PERSONA: DYNAMIC AND COMPOSITIONAL INFERENCE-TIME PERSONALITY CONTROL VIA ACTIVATION VECTOR ALGEBRA

Xiachong Feng[1,2], **Liang Zhao**[1], **Weihong Zhong**[1], **Yichong Huang**[1], **Yuxuan Gu**[1], **Lingpeng Kong**[2†], **Xiaocheng Feng**[1†], **Bing Qin**[1†]
[1]Harbin Institute of Technology [2]The University of Hong Kong
fengxc@hku.hk

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ABSTRACT

Current methods for personality control in Large Language Models rely on static prompting or expensive fine-tuning, failing to capture the dynamic and compositional nature of human traits. We introduce PERSONA, a training-free framework that achieves fine-tuning level performance through direct manipulation of personality vectors in activation space. Our key insight is that personality traits appear as extractable, approximately orthogonal directions in the model's representation space that support algebraic operations. The framework operates through three stages: PERSONA-BASE extracts orthogonal trait vectors via contrastive activation analysis; PERSONA-ALGEBRA enables precise control through vector arithmetic (scalar multiplication for intensity, addition for composition, subtraction for suppression); and PERSONA-FLOW achieves context-aware adaptation by dynamically composing these vectors during inference. On PersonalityBench, our approach achieves a mean score of 9.60, nearly matching the supervised fine-tuning upper bound of 9.61 without any gradient updates. On our proposed PERSONA-EVOLVE benchmark for dynamic personality adaptation, we achieve up to 91% win rates across diverse model families. These results provide evidence that aspects of LLM personality are mathematically tractable, opening new directions for interpretable and efficient behavioral control[1].

1 INTRODUCTION

Personality control in Large Language Models (LLMs) is essential for human-centric applications including healthcare (Guo et al., 2024; He et al., 2025; Ju et al., 2025), education (OpenAI, 2025; Anthropic, 2025), and social simulation (Mou et al., 2024; Feng et al., 2025). In these domains, the personality exhibited by an LLM directly influences user trust, engagement, and decision-making (Dong et al., 2025). However, existing methods fail to achieve both precise control and computational efficiency: prompting (Jiang et al., 2024; Yeo et al., 2025; La Cava & Tagarelli, 2025) suffers from instability and inconsistency, while fine-tuning (Pan & Zeng, 2023; Liu et al., 2024) requires substantial resources for each personality configuration. More fundamentally, both approaches treat personality as static and monolithic, unable to capture the dynamic, compositional nature of human behavioral traits (Chapman et al., 2011).

We introduce PERSONA, a training-free framework that achieves fine-tuning level performance through direct manipulation of personality vectors in activation space. Our key insight is that personality traits appear as extractable, approximately orthogonal directions in the model's representation space that support algebraic operations. This geometric perspective transforms personality control from a problem of text engineering or gradient optimization to one of vector arithmetic in high-dimensional space. Remarkably, without any gradient updates, our approach achieves a mean score of 9.60 on PersonalityBench, nearly matching the supervised fine-tuning upper bound of 9.61.

[†]Corresponding author.
[1]Code and data are publicly available at code

The PERSONA framework operates through three integrated stages. First, PERSONA-BASE extracts orthogonal trait vectors corresponding to the Big Five (OCEAN) model (Roccas et al., 2002) through contrastive activation analysis, providing evidence that personality dimensions are (approximately) linearly encoded in selected activation layers (Turner et al., 2023; Marks & Tegmark, 2024; Rimsky et al., 2024). Second, PERSONA-ALGEBRA demonstrates that these vectors support precise mathematical operations: scalar multiplication controls trait intensity, vector addition enables multi-trait composition, and subtraction allows targeted trait suppression. Third, PERSONA-FLOW achieves context-aware adaptation by dynamically composing these vectors during inference, enabling real-time personality modulation without predefined scripts.

To systematically evaluate dynamic personality control, we introduce PERSONA-EVOLVE, a benchmark comprising 800 multi-turn dialogue scenarios where models must maintain consistent personas while adapting to evolving conversational contexts. Experimental results demonstrate the effectiveness of our approach: on PERSONA-EVOLVE, we achieve up to 91% win rates across diverse model families (Qwen, Llama, Mistral) in trait adherence, role consistency, and response authenticity. On the external PersonalityBench (Deng et al., 2025), our training-free method matches supervised fine-tuning performance while maintaining lower variance (0.74), confirming both effectiveness and stability in diverse conversational contexts.

Our contributions are threefold:

- **PERSONA-BASE**: Personality traits can be represented as approximately orthogonal, extractable vectors in activation space via contrastive analysis, enabling training-free control that matches fine-tuning performance (9.60 vs 9.61 on PersonalityBench).
- **PERSONA-ALGEBRA and PERSONA-FLOW**: Personality vectors support algebraic operations (scaling, addition, subtraction) with predictable outcomes and dynamic context-aware adaptation during inference, achieving up to 91% win rates across model families.
- **PERSONA-EVOLVE**: A comprehensive benchmark of 800 multi-turn dialogue scenarios for evaluating dynamic personality control, assessing trait adherence, role consistency, and response authenticity.

2 THE PERSONA FRAMEWORK

This section presents the PERSONA framework for compositional personality control in LLMs. We introduce four integrated components: PERSONA-BASE (§2.2) extracts orthogonal OCEAN trait vectors; PERSONA-ALGEBRA (§2.3) enables trait composition via vector arithmetic; PERSONA-FLOW (§2.4) performs dynamic inference-time steering; and PERSONA-EVOLVE (§2.5) benchmarks contextual personality adaptation. We begin by providing a comprehensive overview of the framework architecture before examining each component in detail.

2.1 FRAMEWORK OVERVIEW

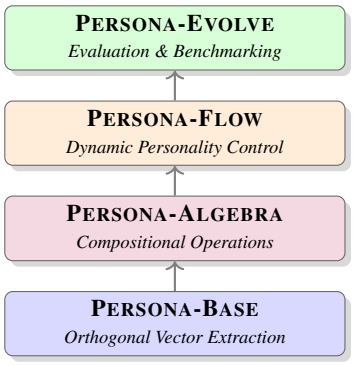

Figure 1: The PERSONA framework.

The PERSONA framework operates through four tightly integrated components, as illustrated in Figure 1. **PERSONA-BASE** extracts orthogonal personality vectors corresponding to the ten poles of the Big Five dimensions from the model's activation space, providing the fundamental building blocks for personality control. **PERSONA-ALGEBRA** leverages these vectors to enable compositional personality manipulation through mathematical operations: scalar multiplication for trait intensity control, vector addition for trait combination, and subtraction for trait suppression. **PERSONA-FLOW** applies these operations dynamically during inference through a predict-then-steer mechanism that adapts personality based on conversational context. **PERSONA-EVOLVE** validates the framework through systematic benchmarks that assess trait expressiveness, role consistency, and personality coherence across diverse scenarios.

2.2 PERSONA-BASE: FOUNDATIONAL PERSONALITY VECTORS

PERSONA-BASE establishes a systematic library of orthogonal personality vectors that serve as fundamental building blocks for personality control. These vectors form the foundation for all algebraic operations and enable modular construction of complex personalities, making their quality critical to the framework's effectiveness.

2.2.1 VECTOR EXTRACTION METHOD

We adopt the OCEAN model, an empirically validated framework in psychology that describes personality through five core dimensions: Openness, Conscientiousness, Extraversion, Agreeableness, and Neuroticism. Each dimension spans a continuum between opposing traits, shown in Table 1. Detailed definitions for each trait pole are provided in Appendix A.3.

Table 1: OCEAN personality vectors with opposing trait poles for each dimension.

Dimension	Abbr.	High Pole	Low Pole
Openness	O	v_Inventive	v_Consistent
Conscientiousness	C	v_Dependable	v_Careless
Extraversion	E	v_Outgoing	v_Solitary
Agreeableness	A	v_Compassionate	v_Self-interested
Neuroticism	N	v_Nervous	v_Calm

To extract personality vectors for the ten trait poles, we employ the automated pipeline from Persona Vectors (Chen et al., 2025a). This method isolates trait representations within the model's activation space through contrastive prompting. The extraction process consists of three stages:

First, a frontier LLM generates necessary artifacts from trait descriptions: (1) five pairs of contrastive system prompts that either elicit or suppress the trait, (2) forty evaluation questions designed to evoke trait-relevant behavior, with half used for extraction and half for validation, and (3) an evaluation rubric for GPT-4.1-mini to score trait expression from 0 to 100. Detailed prompts, examples and generated data are provided in Appendix A.1. The robustness of this approach to the choice of generator LLM is examined in Appendix A.2. Second, the model generates responses to these questions under both positive and negative prompt conditions. During generation, we collect internal residual stream activations from each response. Finally, we compute the persona vector as the difference between mean activations from trait-expressing and -suppressing response groups. This yields a directional vector representing the target trait. Following Chen et al. (2025a), we select vectors from the most effective model layer. The empirical justification for single-layer steering and the selection of the optimal layer is provided in Appendix A.4.

For downstream use, we steer the model by residual addition at the selected layer. Given a persona vector v_l from layer l, we modify the residual stream as $h_l \leftarrow h_l + \alpha\, v_l$, where α is the steering coefficient and h_l denotes the residual activation at layer l. Positive/negative α amplifies/suppresses the associated trait pole; we use this operation consistently in §2.3 and §2.4.

2.2.2 ORTHOGONALITY VALIDATION

To validate the extracted vectors, we evaluate their effectiveness through causal steering (Turner et al., 2023; Rimsky et al., 2024) on the held-out evaluation questions, steering activations as above ($h_l \leftarrow h_l + \alpha\, v_l$).

Table 2 shows each vector reliably induces its corresponding trait, with expression scores increasing monotonically with positive coefficients. Negative coefficients effectively suppress traits, consistent with behavioral disabling methods (Arditi et al., 2024), as reported previously.

We observe asymmetric steering effects for certain traits. Traits aligned with the model's training objectives (e.g., *dependable*) show ceiling effects at baseline, lim-

Table 2: GPT-4.1-mini evaluated trait expression scores (0-100) for responses steered with varying coefficients. Darker shades indicate stronger expression.

Trait	Type	Steering Coefficient (α)			
		-1.0	0.0	$+1.0$	$+2.0$
Inventive	O+	42.3	63.3	88.4	96.1
Consistent	O-	27.2	51.1	69.2	79.5
Dependable	C+	68.9	93.5	93.7	92.7
Careless	C-	0.7	2.8	83.8	96.2
Outgoing	E+	23.2	45.4	85.0	97.7
Solitary	E-	9.2	30.5	46.3	62.4
Compassionate	A+	71.8	90.8	95.9	97.1
Self-interested	A-	4.8	7.7	12.6	20.8
Nervous	N+	6.6	13.0	45.6	96.8
Calm	N-	54.8	96.1	96.6	95.5